



Specialized Avatar Tutors: Personalized Learning with Prerequisite Scaffolding

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Abstract

We present **ZOOAI Avatar Tutors**, a system of domain-specialized, animal-themed AI tutors that provide personalized education through prerequisite-aware scaffolding. Each avatar (Oliver Owl for logic, Diana Dolphin for biology, Sam Sloth for computer science, Elena Elephant for mathematics, Pedro Penguin for physics) maintains a prerequisite knowledge graph $\mathcal{G} = (V, E)$, a per-learner mastery vector $\mu \in [0, 1]^{|V|}$, and a scaffolding policy derived from active inference. The system enforces a mastery threshold of $\theta = 0.8$: no concept is introduced until all prerequisites exceed this threshold. Multi-source retrieval-augmented generation (RAG) with mandatory inline citations ensures factual grounding, while calibrated confidence indicators express honest uncertainty. We formalize the prerequisite scaffolding as a partially observable Markov decision process (POMDP) and prove that the active inference teaching policy converges to the optimal topological ordering of concepts. Controlled experiments show 38% higher learning gains and $2.1\times$ better retention at 4-week follow-up compared to general-purpose LLMs, with 47% improvement in transfer to novel problems.

Keywords: intelligent tutoring systems, prerequisite scaffolding, retrieval-augmented generation, mastery learning, active inference

1 Introduction

General-purpose LLMs fail as tutors in three specific ways. First, they assume prerequisite knowledge that the learner may not have, leading to explanations that are technically correct but incomprehensible. Second, they drift between teaching styles, sometimes explaining patiently and sometimes tersely, breaking the pedagogical relationship. Third, they generate plausible-sounding explanations without citations, undermining the learner’s ability to verify claims and develop critical thinking.

ZOOAI Avatar Tutors address these failures through three mechanisms:

1. **Prerequisite scaffolding:** A knowledge graph tracks concept dependencies. The system assesses mastery before introducing new material and provides remediation for gaps.
2. **Stable personas:** Each avatar has a fixed domain, teaching style, and personality. Oliver Owl always uses Socratic questioning; Sam Sloth always decomposes problems step-by-step.
3. **Multi-source RAG with citations:** Every factual claim is grounded in retrieved passages from vetted sources (textbooks, peer-reviewed papers, educational platforms), with inline citations rendered in the response.

2 Learning Science Foundations

The system is grounded in four established theories.

2.1 Cognitive Load Theory (Sweller, 1988)

Sweller [1] identifies three types of cognitive load: intrinsic (complexity of the material), extraneous (complexity of the presentation), and germane (load devoted to learning). Our system manages intrinsic load through prerequisite sequencing (presenting concepts in order of increasing complexity), reduces extraneous load through consistent avatar interfaces, and optimizes germane load through scaffolded practice.

2.2 Zone of Proximal Development (Vygotsky, 1978)

Vygotsky [2] defines the ZPD as the range between what a learner can do independently and what they can do with guidance. The mastery vector μ approximates the learner’s current capability. The scaffolding policy selects concepts within the ZPD—those whose prerequisites are mastered but which have not yet been mastered themselves.

2.3 Mastery Learning (Bloom, 1968)

Bloom [3] demonstrated that mastery-based instruction (requiring 80%+ proficiency before advancement) dramatically reduces variance in student achievement. We implement this via the mastery threshold $\theta = 0.8$.

2.4 Testing Effect (Roediger & Karpicke, 2006)

Roediger and Karpicke [4] showed that retrieval practice (testing) produces stronger long-term retention than re-study. Avatar tutors embed low-stakes quizzes throughout interactions, using spaced repetition to reinforce key concepts.

3 Formal Framework

3.1 Prerequisite Knowledge Graph

Definition 3.1 (Prerequisite Graph). *A prerequisite graph is a directed acyclic graph $\mathcal{G} = (V, E)$ where V is the set of concepts and $(u, v) \in E$ means concept u is a prerequisite for concept v . We write $\text{prereq}(v) = \{u \mid (u, v) \in E\}$.*

Example 3.1. *In the mathematics domain (Elena Elephant):*

arithmetic $\rightarrow$ algebra $\rightarrow$ calculus $\rightarrow$ differential_equations

algebra $\rightarrow$ linear_algebra $\rightarrow$ eigenvalues

3.2 Mastery Vector

Definition 3.2 (Mastery State). *The mastery state of a learner is a vector $\mu \in [0, 1]^{|V|}$ where μ_v is the estimated probability that the learner has mastered concept v . Mastery is assessed through diagnostic questions.*

The mastery vector is updated via Bayesian inference after each assessment:

$$\mu_v^{(t+1)} = \frac{P(\text{response} \mid \mu_v^{(t)}) \cdot \mu_v^{(t)}}{P(\text{response})} \quad (1)$$

where $P(\text{response} \mid \mu_v^{(t)})$ is a logistic model relating mastery to response probability.

3.3 Scaffolding as a POMDP

Definition 3.3 (Teaching POMDP). *The avatar tutoring problem is a POMDP $(\mathcal{S}, \mathcal{A}, \mathcal{O}, T, O, R, \gamma)$:*

- $\mathcal{S} = [0, 1]^{|V|}$: hidden mastery state.

Avatar	Domain	Style	$ V $	$ E $
Oliver Owl	Logic	Socratic questioning	127	203
Diana Dolphin	Biology	Analogies & examples	248	412
Sam Sloth	CS	Step-by-step decomp.	183	297
Elena Elephant	Math	Systematic building	215	384
Pedro Penguin	Physics	Experiment-driven	196	341

Table 1: Avatar roster with prerequisite graph sizes.

- $\mathcal{A}$: teaching actions (*demonstrate, guided practice, independent practice, remediate, quiz, advance*).
- $\mathcal{O}$: observations (*quiz scores, response quality, time-on-task*).
- $T(s' | s, a)$: mastery transition (*how mastery changes given a teaching action*).
- $O(o | s, a)$: observation model (*how quiz scores relate to mastery*).
- $R(s, a)$: reward (*learning gain per action*).
- γ : discount factor.

3.4 Active Inference Teaching Policy

The avatar selects actions to minimize expected free energy (EFE), following the active inference framework [8]:

$$G(\pi) = \underbrace{-\mathbb{E}_{Q(\tilde{o}|\pi)}[\log P(\tilde{o})]}_{\text{pragmatic: achieve learning goals}} + \underbrace{\mathbb{E}_{Q(\tilde{o}|\pi)}[H[Q(\tilde{s} | \tilde{o}, \pi)]]}_{\text{epistemic: reduce uncertainty about mastery}} \quad (2)$$

The pragmatic term drives the avatar to select actions that lead to observations consistent with the learner’s goals (high mastery on target concepts). The epistemic term drives the avatar to select actions that reduce its uncertainty about the learner’s true mastery state (e.g., administering diagnostic quizzes).

Theorem 3.1 (Topological Teaching Order). *Under the EFE teaching policy with prerequisite graph $\mathcal{G}$ and mastery threshold θ , the sequence of concepts presented converges to a topological ordering of $\mathcal{G}$. Specifically, concept v is never introduced before all $u \in \text{prereq}(v)$ satisfy $\mu_u \geq \theta$.*

Proof. The pragmatic term $-\log P(\tilde{o})$ penalizes observations inconsistent with goals. If the avatar attempts concept v before prerequisites are mastered, the expected observation (quiz score) is low, incurring high pragmatic cost. The EFE-minimizing policy therefore delays v until $\mu_u \geq \theta$ for all $u \in \text{prereq}(v)$. Since $\mathcal{G}$ is a DAG, the resulting sequence is a topological ordering. $\square$

4 Avatar Roster

Each avatar’s teaching style is implemented as a system prompt template combined with a constrained decoding strategy:

Source Type	Count	Quality Gate
Textbooks (OpenStax, etc.)	12–24 per domain	Editorial review
Peer-reviewed papers	PubMed/arXiv	$h\text{-index} \geq 20$
Educational platforms	Khan, MIT OCW	Curated by educators
Reference works	Britannica, Wikipedia	Featured articles only

Table 2: Source types and quality gates.

Listing 1: Avatar persona enforcement

```

class AvatarPersona:
    def __init__(self, avatar_config):
        self.name = avatar_config["name"]
        self.style = avatar_config["teaching_style"]
        self.system_prompt = self.build_system_prompt()

    def build_system_prompt(self):
        return f"""You are {self.name}, a {self.style}.
RULES (never violate):
1. Always cite sources with [N] notation.
2. Never introduce concept X until all
   prerequisites have mastery  $\geq 0.8$ .
3. Express uncertainty honestly: "I'm not
   certain about this, let me check..."
4. Use the I-do/We-do/You-do scaffolding.
5. Maintain your teaching style consistently.
"""

```

5 Multi-Source RAG with Citations

5.1 Source Curation

Each avatar domain has a curated source index:

5.2 Retrieval Pipeline

The RAG pipeline uses dense retrieval (E5-large-v2), cross-encoder re-ranking (MS-MARCO MiniLM), and maximal marginal relevance (MMR) for source diversity:

$$\text{MMR}(d) = \lambda \cdot \text{sim}(d, q) - (1 - \lambda) \cdot \max_{d' \in S} \text{sim}(d, d') \quad (3)$$

where q is the query, S is the already-selected set, and $\lambda = 0.7$ balances relevance against diversity. A minimum of 3 distinct source types is required for any response.

5.3 Citation Format

Listing 2: Citation-enforced generation

```

def generate_with_citations(avatar, query, sources):
    retrieved = avatar.rag.retrieve(query, top_k=10)
    diverse = mmr_select(retrieved, min_types=3)

    prompt = f"""Based_on_these_sources:
{format_sources(diverse)}

Answer_the_student's_question:{query}

CITE_every_factual_claim_with_[N]_where_N_is
the_source_number._If_unsure,_say_so."""

    response = avatar.model.generate(prompt)

    # Verify all facts have citations
    uncited = find_uncited_claims(response)
    if uncited:
        response = avatar.add_missing_citations(
            response, uncited, diverse)

    return response

```

6 Confidence Calibration

Definition 6.1 (Calibrated Confidence). *A tutor’s confidence estimate $c \in [0, 1]$ for a claim is calibrated if:*

$$P(\text{claim is correct} \mid c = p) = p \quad \forall p \in [0, 1]$$

We implement calibration via temperature scaling [13] on the logits of the generating model, trained on a held-out set of factual claims with known ground truth. When confidence falls below 0.7, the avatar explicitly expresses uncertainty:

“I’m not fully certain about this. Let me check my sources... According to [3], the mechanism involves X, but [5] suggests Y may also play a role. Would you like to explore both possibilities?”

7 Scaffolding Algorithm

8 BitDelta Personalization

Each learner’s avatar is personalized via BitDelta [10]:

$$W_{\text{learner}} = W_{\text{avatar_base}} + \gamma_{\text{learner}} \cdot \text{sign}(\delta_{\text{learner}}) \quad (4)$$

The adapter captures the learner’s:

- Mastery levels across all concepts.

Algorithm 1 Prerequisite-Aware Teaching Loop

```
1: Input: Target concept  $v$ , mastery vector  $\mu$ , graph  $\mathcal{G}$ 
2: {Phase 1: Prerequisite assessment}
3: for each  $u \in \text{prereq}(v)$  do
4:   if  $\mu_u < \theta$  then
5:     questions  $\leftarrow$  generate_diagnostic( $u$ )
6:      $\mu_u \leftarrow$  assess(questions)
7:   end if
8: end for
9: {Phase 2: Remediation}
10: gaps  $\leftarrow \{u \in \text{prereq}(v) \mid \mu_u < \theta\}$ 
11: for each  $u \in \text{topological\_sort}(\text{gaps})$  do
12:   I do: Avatar demonstrates  $u$  with citations
13:   We do: Guided practice (3 problems)
14:   You do: Independent practice (3 problems)
15:    $\mu_u \leftarrow$  assess_mastery( $u$ )
16:   if  $\mu_u < \theta$  then
17:     Additional remediation with alternative explanations
18:   end if
19: end for
20: {Phase 3: New concept introduction}
21: if  $\forall u \in \text{prereq}(v) : \mu_u \geq \theta$  then
22:   Teach  $v$  using avatar’s teaching style
23:   Embed retrieval practice (spaced repetition)
24:    $\mu_v \leftarrow$  assess_mastery( $v$ )
25: end if
26: return Updated  $\mu$ 
```

- Preferred explanation style (formal vs. intuitive).
- Pace (how many examples before assessment).
- Common misconceptions (encoded as negative examples in the fine-tuning data).

Storage per learner: 0.7GB (for the 72B Eco-1 base).

9 Experiments

9.1 Controlled Study Design

We conducted a controlled study with 240 participants (university undergraduates) across three conditions:

1. **Avatar Tutor:** ZOOAI avatar with full scaffolding.
2. **General LLM:** GPT-4 with a generic “helpful tutor” system prompt.
3. **Textbook only:** Access to OpenStax textbook chapters.

Topic: introductory linear algebra (matrix operations through eigenvalues). Duration: 4 sessions of 45 minutes each over 2 weeks.

Metric	Avatar	Gen. LLM	Textbook
Normalized learning gain	0.62	0.45	0.31
4-week retention	78%	37%	41%
Transfer accuracy	71%	48%	39%
Factual accuracy of tutor	94%	82%	N/A
Student trust (1–5 Likert)	4.3	3.1	3.8

Table 3: Controlled study results (n=80 per condition).

Configuration	Learning Gain	Dropout Rate
Full scaffolding ($\theta = 0.8$)	0.62	5%
Scaffolding ($\theta = 0.6$)	0.51	12%
No scaffolding	0.39	23%

Table 4: Effect of prerequisite scaffolding threshold.

9.2 Results

Learning gains: 38% higher than general LLM ($p < 0.001$, Cohen’s $d = 0.89$).

Retention: $2.1\times$ better at 4-week follow-up.

Transfer: 47% improvement on novel problems requiring prerequisite integration.

Factual accuracy: The mandatory citation requirement raised factual accuracy from 82% (general LLM) to 94%.

9.3 Prerequisite Scaffolding Impact

The 0.8 threshold produces the highest learning gains and lowest dropout. Lower thresholds allow learners to advance before mastering prerequisites, leading to confusion and disengagement.

10 Related Work

Intelligent tutoring systems. AutoTutor [5] uses expectation-misconception tailored dialogue. Cognitive Tutors [6] use model tracing for just-in-time hints. ZOOAI avatars combine both approaches with modern LLMs and RAG.

AI in education. Khan Academy’s Khanmigo uses GPT-4 for tutoring but lacks prerequisite tracking and stable personas. Duolingo’s AI features focus on language learning without knowledge graph structure.

Pedagogical agents. Lester et al. [7] demonstrated the “persona effect”: animated agents increase engagement. ZOOAI avatars leverage this effect through animal-themed characters.

RAG for education. Borgeaud et al. [11] and Izacard et al. [12] demonstrate retrieval-augmented generation but do not enforce citation or source diversity requirements.

11 Conclusion

ZOOAI Avatar Tutors demonstrate that domain-specialized AI tutors with prerequisite scaffolding, stable personas, and citation-enforced RAG substantially outperform general-purpose LLMs as ed-

educational tools. The formal framework—prerequisite graphs, mastery vectors, and active inference teaching policies—provides theoretical guarantees on teaching order (Theorem 3.1) and connects to the broader Zoo AI architecture through BitDelta personalization and Eco-1 behavioral experts. The 38% improvement in learning gains and $2.1\times$ retention improvement validate the approach for deployment in Zoo’s educational ecosystem.

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